Benchmark Requirements

Domain: Overleaf / LaTeX

Task: Develop a structured LaTeX research document on factors affecting student academic performance.

Required Project Content

- Use a 60-observation synthetic student-performance dataset.
- Include chapters for Introduction, Literature Review, Methodology, Results, and Conclusion.
- Use multiple linear regression to examine academic performance.
- Include descriptive statistics and Pearson correlation analysis with visible calculations where appropriate.
- Include the regression model, coefficients, model performance measures, a worked prediction, residual-based validation, and interpretation.
- Include practical recommendations and clearly state the limitations arising from the synthetic dataset.

LaTeX Requirements

- Keep the project editable as LaTeX source files.
- Connect all chapter files correctly through the main document.
- Compile the complete document successfully in Overleaf.
- Use existing references consistently; add new references only when necessary.

Expected Final State

- A complete, coherent research document containing all five chapters.
- Correct and traceable statistical results based on the supplied synthetic dataset.
- A successfully compiled final PDF together with the editable project source.
- A project suitable for incremental benchmark review through its milestone history.

Important: The synthetic dataset and resulting model are demonstrative and should not be presented as evidence from a real student population.